FINAL PROJECT REPORT

MyTrip

Budget Road Trip Planner

CS 4398

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Summer 2026

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1. INTRODUCTION

1.1 Document Purpose

The purpose of this report is to describe the requirements and main functions of MyTrip – Budget Road Trip Planner. The report explains what the website is designed to do, how users interact with it, and how the system will be tested.
This document will also describe the user interface, use cases, functional requirements, non-functional requirements, and expected behavior of the system.

1.2 Product Scope

MyTrip is a web-based travel planning application designed to help users organize affordable road trips. The website combines important trip-planning information in one place so that users do not have to search through several different websites.
The system allows users to create and manage road trips, enter a starting location and destination, and display the travel route between the selected locations. Users can search for affordable hotels near the destination or along the travel route and locate nearby gas stations.
MyTrip also estimates fuel costs based on the trip distance, calculates the estimated total cost of the trip, and compares the estimated expenses with the user's budget. Users can save and edit existing trip plans and view a summary of the completed travel plan.
The main goal of MyTrip is to make road-trip planning easier and help travelers make better decisions based on their expected travel expenses.

1.3 Intended Audience

MyTrip is intended for individuals and families who are planning road trips and want to organize their travel information in one place.
The website is designed for users with basic experience using a web browser. No special technical knowledge or travel-planning experience is required. Users should be able to enter their trip information, review the route, search for hotels and gas stations, estimate travel costs, and compare the estimated cost with their budget.

2. OVERALL DESCRIPTION

2.1 Product Perspective

MyTrip is a web-based budget travel planning application. It is designed to bring the main parts of road-trip planning into one website.
Normally, travelers may need to use different websites to search for routes, hotels, gas stations, and estimated travel expenses. MyTrip provides a more organized way to review this information and prepare a trip plan.
The application can be accessed through a modern web browser with an active internet connection. Users do not need to install special software on their computers.

2.2 Product Functions

The MyTrip application provides the following major functions:
• Create and manage road trips.
• Enter a starting location and destination.
• Display the travel route between the selected locations.
• Search for affordable hotels near the destination or along the travel route.
• Search for nearby gas stations.
• Estimate fuel costs based on the trip distance.
• Calculate the estimated total cost of the trip.
• Compare the estimated trip expenses with the user’s budget.
• Save and edit existing trip plans.
• Display a summary of the completed travel plan.

These functions help the user organize important route and cost information before completing a road-trip plan.

2.3 User Characteristics

The intended users of MyTrip are individuals and families planning personal road trips. Users are expected to have basic computer skills and basic experience using a web browser. No technical knowledge or specialized training is required to use the website. The interface should be simple enough for a first-time user to enter trip information, review travel options, and understand the estimated cost of the trip.

2.4 Operating Environment

MyTrip is designed to work through commonly used web browsers, including:

• Google Chrome
• Microsoft Edge
• Mozilla Firefox
• Safari

The website can be accessed using a desktop computer, laptop, or supported tablet. A stable internet connection is required to retrieve route, hotel, and gas-station information.

3. REQUIREMENTS

3.1 Functional Requirements

The functional requirements describe the main actions that the MyTrip website must perform.

FR-01: Create a Road Trip
The system shall allow the user to create a new road-trip plan.
FR-02: Manage Road Trips
The system shall allow the user to view and manage existing road-trip plans.
FR-03: Enter Trip Locations
The system shall allow the user to enter a starting location and destination.
FR-04: Validate Trip Locations
The system shall require both a starting location and destination before generating a route.
FR-05: Display Travel Route
The system shall display the travel route between the selected starting location and destination.
FR-06: Search for Hotels
The system shall allow the user to search for affordable hotels near the destination or along the travel route.
FR-07: Display Hotel Results
The system shall display available hotel options related to the selected destination or route.
FR-08: Search for Gas Stations
The system shall allow the user to search for nearby gas stations along the travel route.
FR-09: Display Gas Station Results
The system shall display available gas stations related to the selected route.
FR-10: Estimate Fuel Cost
The system shall estimate the fuel cost of the trip based on the travel distance.

FR-11: Enter a Trip Budget
The system shall allow the user to enter a budget for the planned trip.
FR-12: Calculate Total Trip Cost
The system shall calculate and display the estimated total cost of the trip.
FR-13: Compare Cost with Budget
The system shall compare the estimated trip cost with the budget entered by the user.
FR-14: Display Budget Status
The system shall indicate whether the estimated trip cost is within or above the user's budget.
FR-15: Save a Trip Plan
The system shall allow the user to save a trip plan.
FR-16: Open a Saved Trip Plan
The system shall allow the user to open and review a previously saved trip plan.
FR-17: Edit a Trip Plan
The system shall allow the user to edit information in a saved trip plan.
FR-18: Update Trip Results
The system shall update the route or estimated cost when the user changes information that affects the trip plan.
FR-19: Display Trip Summary
The system shall display a summary of the completed travel plan, including the starting location, destination, route, and estimated trip expenses.
FR-20: Handle Missing Results
The system shall inform the user when a route, hotel, or gas-station result cannot be found.

3.2 Non-Functional Requirements

The non-functional requirements describe the expected quality and operation of the MyTrip website.

Reliability
NFR-01: The website shall operate correctly when a stable internet connection and required online services are available.
NFR-02: The system shall return available route, distance, hotel, and gas-station information using the configured online services.

Robustness
NFR-03: The system shall display a clear error message when required information is missing or invalid.

NFR-04: The system shall display an appropriate message when the internet connection or an external service is unavailable.
NFR-05: The website shall not crash when a route, hotel, or gas-station search returns no results.

Performance
NFR-06: Route generation should complete within 10 seconds under normal network conditions.
NFR-07: Website pages and search results should load within a reasonable amount of time under normal use.

Maintainability
NFR-08: The source code shall follow a clear object-oriented structure.
NFR-09: The system shall use the Model-View-Controller architecture to separate the application logic, interface, and user actions.
NFR-10: The source code shall be managed using Git to support collaboration and future updates.

Security
NFR-11: The system shall protect saved trip information from unauthorized modification.
NFR-12: Information entered by the user shall be handled only for the purpose of creating and managing the trip plan.

Usability
NFR-13: The website shall provide a simple and understandable interface.
NFR-14: Buttons, input fields, search results, and error messages shall be clearly labeled.
NFR-15: A first-time user should be able to create a basic trip plan and view an estimated cost without specialized training.

Portability
NFR-16: The website shall work through commonly used modern web browsers.
NFR-17: The user shall not be required to install special desktop software to access the application.

4. USE CASES

The following use cases describe how travelers and administrators interact with the main functions of the MyTrip website.

4.1 Use Case: Manage Users

Name	MANAGE USERS
Primary Actor:	Admin
Brief Description	The administrator reviews and manages traveler and vendor accounts registered on the website.
Pre-Conditions	The administrator is logged in and has an active internet connection.
Post-Conditions	Admin is shown on a list of registered travelers and vendors and can approve, suspend, or remove accounts.
Main Flow:	1. The administrator logs in to the system. 2. The administrator opens the list of registered users. 3. The administrator reviews traveler and vendor accounts. 4. The administrator approves, edits, suspends, or removes an account as needed.
Alternate Flow / Exceptions	If the internet connection is unavailable, the system displays an error message. If no users are registered, the system displays an empty list.

4.2 Use Case: Plan Trip Route

Name	PLAN TRIP ROUTE
Primary Actor:	Traveler

Brief Description	The traveler enters the trip locations and generates a route for the planned road trip.
Pre-Conditions	Traveler is logged in with an active internet connection.
Post-Conditions	The system displays a map with the generated route and suggested stops along the way.
Main Flow:	1. Traveler logs in and enters start location, destination, and dates. 2. Traveler clicks Find Route. 3. The system displays the route on a map with suggested stops.
Alternate Flow / Exceptions	If the internet is down, an error page is displayed. If no route can be found, the traveler is prompted to revise the locations entered.

4.3 Use Case: Calculating Travel Expenses

Name	Calculating Travel Expenses
Primary Actor:	Traveler
Brief Description	Application/site will calculate and sum travel expenses based off users preferences and trip.
Pre-Conditions	Traveler states destination and budget
Post-Conditions	The system finds location and suggests itemized activities and lodging within travelers budget.
Main Flow:	1.Traveler logs in and enters location, budget and dates. 2. System finds activities related to location and within budget 3. Travelers select activities and lodges they like 4. System sums and suggest more.

Alternate Flow / Exceptions	If the internet is down, an error page is displayed. If no activities can be found, the traveler is prompted to revise the locations entered. The system might suggest out of budget activities and lodging if too few results were shown.

4.4 Case: Search for Nearby Gas Stations

Name	SEARCH FOR NEARBY GAS STATIONS
Primary Actor:	Traveler
Brief Description	The traveler searches for gas stations near the selected travel route.
Pre-Conditions	A valid travel route has been generated.
Post-Conditions	The nearby gas-station results are displayed.
Main Flow:	1. The traveler views the generated route. 2. The traveler selects the gas-station search option. 3. The system searches for nearby gas stations. 4. The system displays the available gas-station results.
Alternate Flow / Exceptions	If no gas stations are found, the system informs the traveler that no results are available for the selected route.

4.5 Use Case: Search for Affordable Hotels

Name	SEARCH FOR AFFORDABLE HOTELS
Primary Actor:	Traveler
Brief Description	The traveler searches for affordable hotels near the destination or along the selected travel route.

Pre-Conditions	A valid travel route has been generated.
Post-Conditions	The available hotel options are displayed for the traveler.
Main Flow:	1. The traveler views the generated route. 2. The traveler selects the hotel-search option. 3. The system searches for hotels near the destination or route. 4. The system displays the available hotel results. 5. The traveler reviews the hotel options.
Alternate Flow / Exceptions	If no hotel results are found, the system informs the traveler that no hotels are currently available for the selected location.

4.6 Use Case: Save and Edit Trip Plans

Name	SAVE AND EDIT TRIP PLANS
Primary Actor:	Traveler
Brief Description	The traveler saves a trip plan and may open it later to review or edit its information.
Pre-Conditions	A trip plan has been created.
Post-Conditions	The trip plan is saved with the latest information.
Main Flow:	1. The traveler selects the option to save the trip. 2. The system saves the trip information. 3. The traveler opens a previously saved trip. 4. The system displays the saved trip information. 5. The traveler changes the desired trip information. 6. The system updates the trip plan. 7. The traveler saves the updated trip.

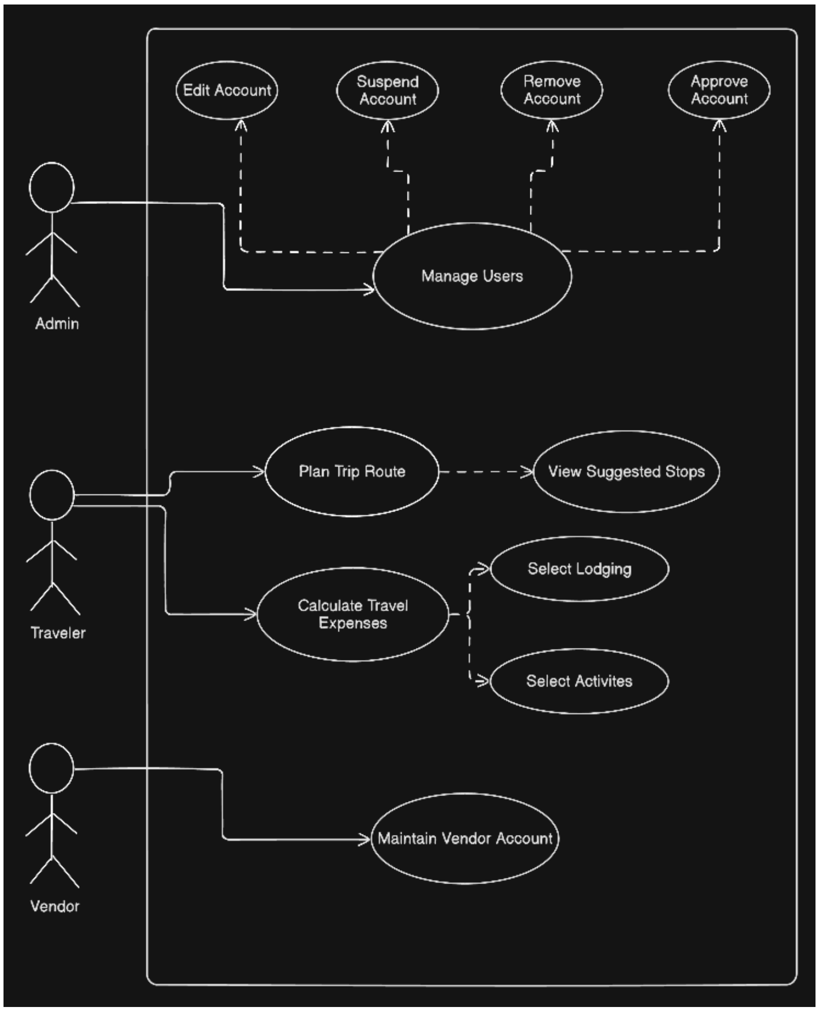
Alternate Flow / Exceptions	If the trip cannot be saved or opened, the system displays an appropriate error message.

4.7 Use Case: View Trip Summary

Name	VIEW TRIP SUMMARY
Primary Actor:	Traveler
Brief Description	The traveler reviews a summary of the completed road-trip plan.
Pre-Conditions	The route and estimated travel expenses have been calculated.
Post-Conditions	The traveler can review the main details of the completed travel plan.
Main Flow:	1. The traveler opens the trip summary. 2. The system displays the starting location and destination. 3. The system displays the selected travel route. 4. The system displays the hotel and gas-station information. 5. The system displays the estimated fuel cost. 6. The system displays the estimated total trip cost. 7. The system displays the budget comparison. 8. The traveler reviews the completed trip plan.
Alternate Flow / Exceptions	If some trip information is unavailable, the system displays the available information and identifies the missing results.

4.8 Use Case Diagram

The use case diagram provides a visual representation of the main actors and their interactions with the MyTrip system.



5. USER INTERFACE

5.1 User Interface Overview

MyTrip provides a simple web-based interface that allows users to plan and review a road trip. The interface is designed to be clear and easy to use for individuals with basic experience using a web browser.
The main pages of the website are the Home Page, Trip Planning Page, Search Results Page, and Trip Summary Page. Each page supports a different part of the trip-planning process.
Users interact with the website by entering information into text fields, selecting buttons, reviewing search results, and navigating between pages.

5.2 Home Page

The Home Page is the first page displayed when the user opens the MyTrip website.
The page should include:
• The MyTrip name and project title.
• A short description of the website.
• A field for entering the starting location.
• A field for entering the destination.
• A button that allows the user to begin planning the trip.
The Home Page should have a simple layout that allows the user to understand the purpose of the website and begin a trip plan without unnecessary steps.

5.3 Trip Planning Page

The Trip Planning Page allows the user to enter and review the main information required for the road trip.

The page should include:
• Starting-location information.
• Destination information.
• Trip-budget information.
• An option to generate or view the travel route.
• An option to search for hotels.
• An option to search for nearby gas stations.

• Navigation buttons that allow the user to continue or return to the previous page.
The system should notify the user when required information is missing or when the entered locations cannot be used to generate a route.

5.4 Search Results Page

The Search Results Page displays information returned for the selected trip.

The page should include:
• The travel route between the starting location and destination.
• Available hotel options near the destination or along the route.
• Nearby gas-station results.
• Estimated fuel-cost information.
• Clear messages when no results are available.

The results should be organized so that the user can easily review the available travel information.

5.5 Trip Summary Page

The Trip Summary Page displays the main details of the completed travel plan.

The page should include:

• Starting location.
• Destination.
• Travel-route information.
• Hotel information.
• Gas-station information.
• Estimated fuel cost.
• Estimated total trip cost.
• Comparison between the estimated trip cost and the user's budget.

The user should be able to review the trip information and return to edit the plan when necessary.

5.6 User Interface Requirements

UI-01: The website shall provide a clear and consistent layout on all pages.
UI-02: The Home Page shall display the MyTrip name and allow the user to begin planning a trip.
UI-03: The starting-location and destination fields shall be clearly labeled.
UI-04: The website shall identify required fields before processing a trip request.
UI-05: Buttons shall use clear labels that describe their actions.
UI-06: The website shall display route, hotel, gas-station, and cost information in an organized format.
UI-07: The website shall display understandable error messages when information is missing, invalid, or unavailable.
UI-08: The user shall be able to move between the main pages without restarting the entire trip plan.
UI-09: The Trip Summary Page shall present the main trip information in one location.
UI-10: The interface shall be usable through commonly supported modern web browsers.

7. AI USE AND PROMPTS

- “Clean up code and ui look but do not change any substantial part of the design or function”
- “Help me make the route option page render accurate results guest can use to plan trips further based on their needs and route.”

8. CONCLUSION

MyTrip – Budget Road Trip Planner is a web-based application designed to make road-trip planning simpler and more organized. The website brings together route planning, hotel searching, gas-station searching, fuel-cost estimation, budget comparison, and trip management in one system. The application allows users to enter a starting location and destination, display a travel route, review affordable hotels and nearby gas stations, estimate fuel expenses, and calculate the estimated total cost of a trip. Users can compare the estimated expenses with their selected budget, save or edit their trip plans, and review a summary of the completed travel plan. The MyTrip project demonstrates how a website can organize important travel information and help users make better budget-conscious decisions before starting a road trip. The system was designed to be understandable, useful, and accessible through commonly used web browsers. Overall, MyTrip meets its main purpose by providing users with a convenient way to plan road trips and review important route and cost information in one place.